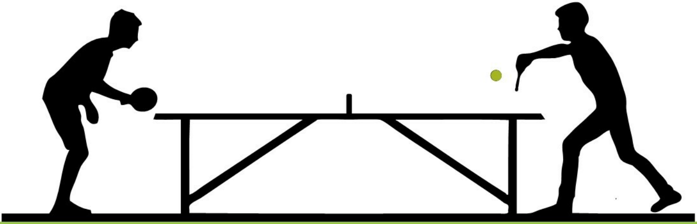

Building the PONG GAME in APP INVENTOR 2

Let's embark on the exciting adventure of building a game which actually works! Let loose your creative instincts and join the author in building the game of Pong on your Android device.

If you were born in the 90s, then you must have played the pong game a lot. It is a simple two-dimensional arcade mode game, which replicates real life table tennis. The objective of the game is to strike the ball towards the opponent in such a way that the latter misses it, enabling you to score a point. Since its inception, it has been a very popular arcade game, with an element of artificial intelligence so that you could play it against the computer.

The concept of animation in App Inventor was introduced in our last article and we have demonstrated how a ball can be moved on the screen using the controls. Taking that concept forward, let's now build a complete game from scratch. So without further delay, let's get into the nitty gritty of building a complete 2D animation game.

The theme of the application

As already described, we will build a pong game for an Android mobile phone using App Inventor 2. The player's objective is to hit the ball using the paddle. Here, we will make a slight modification from the conventional game. We will control the paddle at both ends, so that our objective will be to keep the ball between the paddles. Any failure in keeping the ball within the limit will cause us to lose points. As long as we keep the ball in play, our score will keep increasing.

In the last application, you got to know about the various options and features available on the designer or block editor in the App Inventor browser application. We will use that knowledge here along with a few more components.

GUI requirements

For every application, a graphical user interface or GUI helps the user to interact with the on-screen components. How each component responds to user actions is defined in the block editor section.

As per our requirements, we will need the following components:

1. *Canvas:* The canvas component works as the base for the animation, and it gives a platform over which the object will move according to instructions.
2. *Labels:* These are the static text components used to display some headings or markings on the screen.
3. *Buttons:* These let you trigger the event and are very essential components.
4. *Ball:* This is the in-built component in your App Inventor palette, which will display a ball on the screen. Using its properties, you can set the colour, diameter and various other features of the ball.
5. *Image sprite:* This is a graphical component, which is the same as the ball component, with the addition that you can assign to it any image to make better graphics and animation. Using the properties of the image sprite component, you can set the colour, diameter and various other features of the image sprite.
6. *Clock:* This component is responsible for all the timely events.